Vietnam National University, Ho Chi Minh City
University of Technology
Faculty of Computer Science and Engineering

INTRODUCTION TO ARTIFICIAL INTELLIGENCE (CO3061)

Assignment

Game playing with Search Algorithms

(UPDATE March 9th, 2026)

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Ho Chi Minh City, March 2026

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1 Objectives

- Implement a game-playing agent for a chosen adversarial game.
- Improve programming and problem-solving skills.

2 Requirements and Evaluation

Each group selects a non-trivial game with a large state space (e.g., Chinese Chess, Chess, etc.). Specific requirements:

1. Agent must play according to the rules.
2. Must defeat a random rule-based agent (10/10), meaning in 10 matches regardless of playing first or second, the agent must win.
3. Agent(s) should have different skill levels (poor – average – good, possibly divided into levels from 1 to 10), and methods to evaluate the playing ability of agent(s) should be explored.
4. Only implement Searching algorithms (no Machine Learning) in this assignment.
5. Create a short presentation video (5–10 minutes) about the game and the group's implementation.

3 Submission

- Only one member of each group submits on BKEL.
- Submit a **.zip** file named after the members' studentID (i.e. ID1_ID2_ID3_ID4.zip) containing:
  - Source code files
  - Report file
  - A text file with the video link (YouTube or Google Drive)
- Deadline: **23:55, April 12, 2026**

4 References

- Course materials.
- Minimax algorithm.
- Alpha-beta pruning.
- Monte Carlo Tree Search.

5 Academic Integrity

Handled according to university regulations.